Familiarisation with TPS

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Medical Physics Laboratory II

Name: Souvik Das

Roll No: 241126007

National Institute of Science Education and Research
Jatani, Odisha, India
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Contents

1 Objective

1. Create a radiotherapy treatment plan using a computerised TPS (Treatment Planning System).
2. Perform treatment planning using various modes (e.g., photons, protons, etc.) of treatment and using different setups of beams to find the best treatment plan.

2 Apparatus

- A computer with MatRad software.
- DICOM image of the Patient.

matRad is an open-source treatment planning system (TPS) research and education toolkit developed in MATLAB for external beam radiotherapy planning [1]. It supports the complete basic planning workflow, including DICOM data import, structure handling, dose calculation, inverse planning, and plan evaluation in a graphical user interface. A major strength of matRad is that it allows planning for different modalities such as photons, scanned protons, and scanned carbon ions within a common framework, which makes it useful for comparative learning.

In this experiment, matRad was used as the primary software platform to understand TPS operation, starting from patient image loading to beam setup and optimization. The software enables users to visualize target volume coverage and dose to organs at risk (OARs), and to assess plan quality through dose distributions and dose-volume histograms (DVHs). Because the code is open and modular, it is particularly suitable for academic use, where both clinical concepts and underlying computational methods can be studied together [1].

3 Theory

The most common type of radiation therapy is external beam therapy, which is characterised by directing ionizing radiation from the outside into the patient's body. Treatment planning is an integral part of radiation therapy treatment, performed before the actual patient treatment using specialized software. The treatment planning process aims to simulate the dose deposition within the patient's body and to optimize the dose distribution according to clinical objectives and constraints that define a trade-off between tumour irradiation and normal tissue sparing.

3.1 Treatment Planning Process

The treatment planning process involves several steps, which are outlined below in Figure 1.

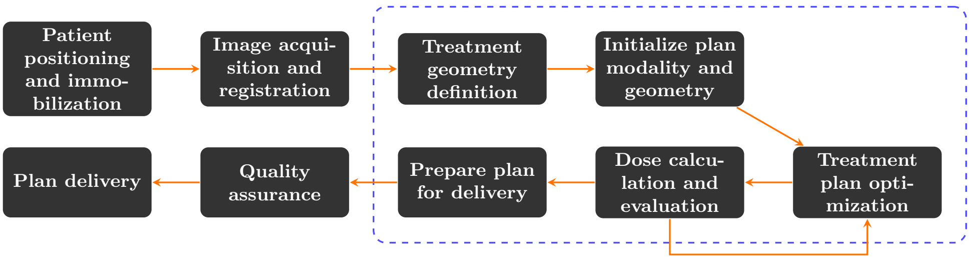


Figure 1: The treatment planning workflow. The dotted blue region represents the TPS-internal steps covered in this experiment.

3.1.1 Treatment Plan Prescribing and reporting

In external photon beam radiotherapy, prescription and reporting require clear definition of target-related volumes and dose-reporting quantities so that treatment intent is unambiguous and reproducible [2].

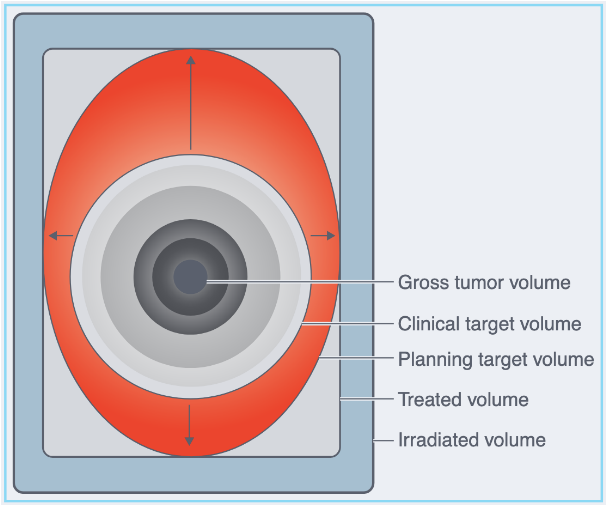


Figure 2: ICRU target and dose specification concepts used in treatment plan prescribing and reporting.

- **Gross Tumor Volume (GTV):** The gross demonstrable extent of malignant disease identified clinically or through imaging.
- **Clinical Target Volume (CTV):** GTV plus regions suspected to contain microscopic disease; it represents the true tissue volume that must receive adequate dose.
- **Internal Target Volume (ITV):** The volume obtained by adding an internal margin to CTV to account for internal physiological motion and shape/position variation.
- **Planning Target Volume (PTV):** CTV (or ITV) plus setup margin to account for patient positioning and machine-related geometric uncertainties during treatment delivery.
- **Organ at Risk Volume (PRV / OAR volume):** A margin-expanded planning representation of an organ at risk, used to better protect normal tissues under motion and setup uncertainty.
- **Irradiated Volume:** The tissue volume receiving a clinically significant dose level (commonly a threshold relative to prescription dose), usually larger than treated volume.
- **Maximum Target Dose:** The highest absorbed dose within the target region that is considered clinically meaningful.
- **Minimum Target Dose:** The lowest absorbed dose found within the target region.
- **Mean Target Dose:** The arithmetic average dose in the target volume, typically calculated over sampled dose points/voxels.
- **Median Target Dose:** The central dose value within the target dose distribution (50th percentile).
- **Modal Target Dose:** The most frequently occurring dose value in the target dose distribution.
- **Hot Spots:** A hot spot is an area outside the target that receives a higher dose than the specified target dose. Like the maximum target dose, a hot spot is considered clinically meaningful only if it covers an area of at least 2 cm^2.

These definitions provide a standard framework for prescription, comparison, and reporting of treatment plans across institutions [2].

3.1.2 Treatment plan and geometry definition

After prescription, treatment planning begins with defining target volumes and critical organs using CT, MRI, or PET imaging. This 3D delineation process, known as contouring or segmentation, identifies structural borders which can be digitally represented as binary masks or triangular meshes. While manual or semi-automated thresholding techniques have traditionally been used, modern workflows increasingly rely on automated segmentation tools to improve efficiency and consistency. These advanced tools encompass atlas-based segmentation—utilizing deformable image registration from a library of clinical cases—as well as artificial intelligence (AI) methods based on convolutional neural networks (CNNs). Despite these advancements, all automated segmentations necessitate thorough manual review and clinical editing to ensure precision before treatment delivery.

3.1.3 Creation of a treatment plan

Once the segmentation of the target and critical organ volumes are available, the 3-D geometry of the treatment planning problem has been defined treatment planning involves the creation of beam, geometry, and reducing output to meet the dosimetric symmetric goals of the segmented geometry, using available option in the treatment delivery device.

3.1.4 Dose display and analysis tools

1. **Isodose curves** The most common way to display the dose distribution is with isodose curves. When points of equal dose are joined together, an isodose curve is produced. These are the curves used for visualizing dose distributions on cross-sectional planes of the patient. This is necessary because the dose to many points in a patient needs to be assessed and compared with the dose at a reference point (usually dmax). A series of isodose curves for a range of different doses, such as %D of 90,70, 50, 30, and 10%, can be combined to form an isodose chart or isodose distribution. This form of display is convenient, since the distribution gives a map of the dose contours in one plane.

2. **Dose distribution in axial, sagittal, and coronal planes(3D dose distribution)** Since patient anatomy is three-dimensional, evaluating dose distribution on a single 2D plane is insufficient. Modern treatment planning systems allow the visualization of isodose distributions across multiple planar views—axial (transverse), sagittal, and coronal. This multi-planar reconstruction (MPR) enables clinicians to assess the volumetric conformity of the dose throughout the target volume and ensure adequate sparing of organs at risk in all spatial dimensions. By scrolling through these contiguous slices, a comprehensive understanding of the 3D dose cloud relative to the patient's anatomy is achieved.

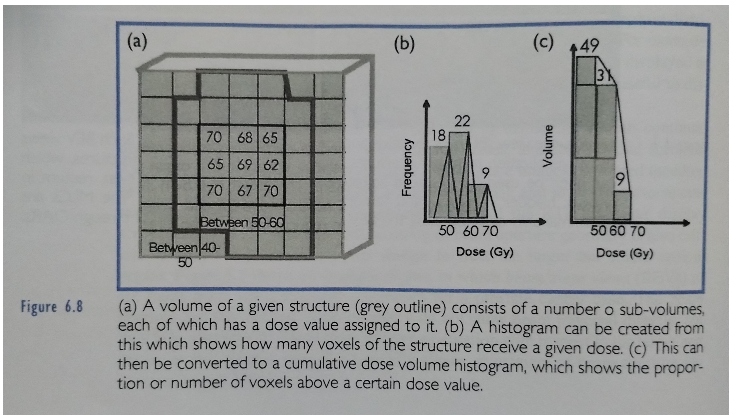


Figure 3: Taken from [3]

3. **Dose-volume histograms (DVHs)** Visual examination of dose distribution data with respect to anatomy allows the physician to evaluate a plan subjectively. Additional tools are required for further quantitative assessment of plans do volume histograms, for example can graphically summarize the large amount of three tools distribution information throughout the volume of the tumor or a normal anatomic structure. The DVD tool combined stepping through the two iso calves are the most common tools used for dose assessment in the clinic. Differential DVH are constructed by grouping the frequency of the voxels within a dose distribution into discrete ranges. Integral or cumulative dose-volume histograms, which are the most commonly used, then take these frequency distributions and group them as a cumulative dose more than a particular dose range shown in Figure 3.

3.1.5 Serial and parallel organs

Normal tissues respond differently to partial-volume irradiation, and this behavior is commonly described using the concepts of serial and parallel organs.

Serial organs: In serial organs, function depends on the integrity of the entire pathway, so high dose to even a small critical sub-volume can cause significant functional loss. Therefore, treatment planning for serial organs is driven mainly by limiting near-maximum dose (for example, metrics such as $D_{0.1\,\mathrm{cc}}$, $D_{1\,\mathrm{cc}}$, or $D_{\max}$ depending on protocol). Typical examples include spinal cord, brainstem, optic chiasm, and major bile ducts.

Parallel organs: In parallel organs, function is distributed across many functional subunits, so damage to part of the organ may be tolerated if sufficient volume is spared. For these organs, planning emphasizes mean dose and volume-based constraints (for example, V_{20}, V_{30}, or mean dose limits). Typical examples include lungs, liver, kidneys, and parotid glands.

In DVH interpretation, serial organs are evaluated with emphasis on the high-dose tail (hot spots), while parallel organs are evaluated with emphasis on how much organ volume receives intermediate to high dose. This distinction is essential for balancing tumour coverage with normal tissue complication risk.

3.1.6 Intensity modulation

1. **Forward and Inverse Planning**: In *forward planning*, the treatment planner places beams into a radiotherapy treatment planning system that can deliver sufficient radiation to a tumor while both sparing critical organs and minimizing the dose to healthy tissue. The planner takes a trial and error approach while using past experience to alter beam parameters, such as beam weights and beam angles, to achieve an acceptable dose distribution. This is achievable when there is little geometric complexity with respect to adjacent critical organs or where doses delivered may be well below dose constraints for critical organs such as in palliative treatments. The treatment planner has the capability of modifying aperture shapes (such as defined by MLCs), beam weights, angles, and dose rates; such a large number of variables rapidly begins to exceed the capabilities of human comprehension. Hence, the concept of *inverse planning* was developed, in which a mathematical optimization of beam delivery parameters is performed to achieve a treatment plan that meets desired dosimetric goals. The required dose goals or objectives are typically provided by the operator in the form of dose threshold points on a DVH curve. These may be simple minimum and maximum doses to structures, or relative organ volumes receiving a given dose. In its simplest form, inverse planning can be used to determine optimal beam weights for a multi-field 3DCRT treatment plan. Inverse planning is typically used to optimize beam delivery components, such as MLC segment shapes and weights, 2D incident fluence (later converted to something deliverable by an MLC), gantry rotation speed, and dose rate.

2. **The rationale for intensity modulation**: The primary rationale for intensity modulation arises from the vital clinical need to highly conform the prescribed dose to complex target volumes, particularly when sensitive organs are adjacent to or enveloped by concave tumor surfaces (e.g., a tumor wrapping around the spinal cord).

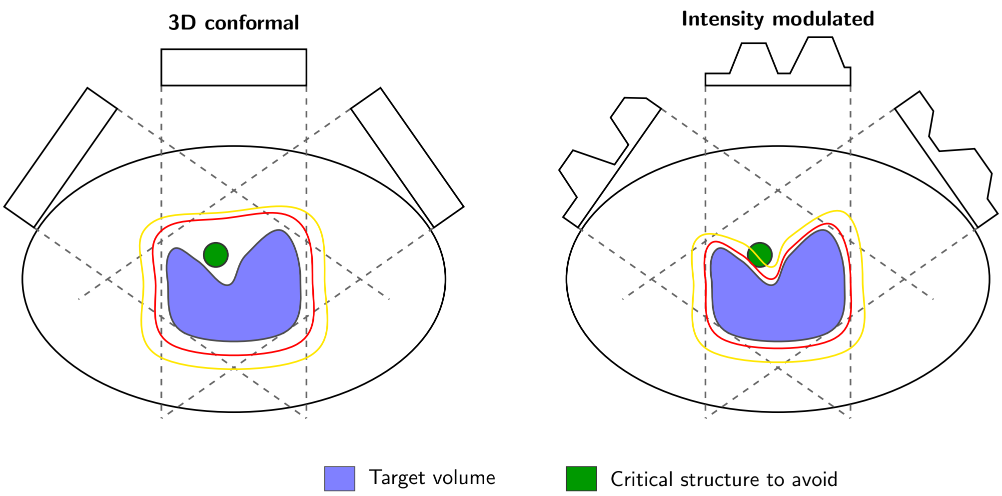


Figure 4: Conformity of dose to a target volume with a concave surface cannot be achieved via simple beam's-eye view planning with unmodulated coplanar beams (left). At right, intensity-modulated beams conform the dose distribution exactly to the target, successfully sparing the critical structure.

While conventional 3D conformal radiotherapy (3DCRT) shapes beam apertures using a 2D beam's-eye view (BEV), it fails to create conformal 3D dose distributions for hollow or cup-shaped targets because unmodulated uniform beams cannot selectively spare the nested critical structures. To achieve true volumetric conformity, non-uniform 2D beam intensity profiles must be generated. This typically involves a two-step planning process: determining suitable beam intensity maps through an inverse planning optimization algorithm, and then physically realizing these intensity profiles using computerized beam delivery mechanisms such as multileaf collimators (MLCs). Occasionally, these steps are combined via direct aperture optimization to overcome the mechanical limitations of the delivery device.

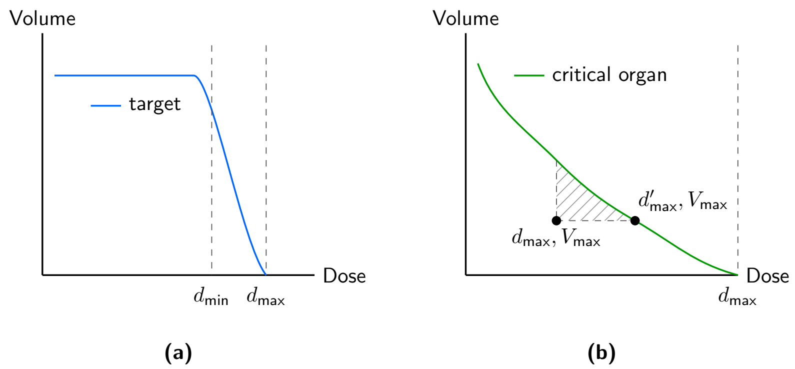


Figure 5: (a) Maximum and minimum target dose goals in terms of a DVH display. Any voxel with dose above the maximum and below the minimum will be penalized. (b) Maximum critical organ dose goal and a critical organ DVH goal. For the DVH goal we want the DVH curve to ideally reach that point. If the DVH passes underneath the point, that DVH goal is considered to be met and does not contribute to the optimization.

3. **Simple optimization mathematics**

Consider the schematic shown in Figure 6. If the matrix containing all beamlet weights is w and the matrix of calculated doses in the patient is d, the dose distribution is obtained from the matrix multiplication $d = Dw$ where D gives the change in d for a unit increment in each beam weight. Note that d and w are single-column matrices and D is a rectangular matrix with the same number of rows as d and the same number of columns as w has rows. In fact, d has as many rows as there are dose calculation points and w has as many rows as there are beamlets in the problem. Figure 6 shows a schematic of an intensity profile (the weights included in the w vector) and a dose distribution described by the d vector. An arrow is drawn (representing an element of the D matrix) indicating the influence of a beam element on a dose element.

Ideally, we would simply invert this matrix algebra expression and obtain w from $w = D^{-1}d$, but this is not a straightforward procedure for reasons such as those discussed above.

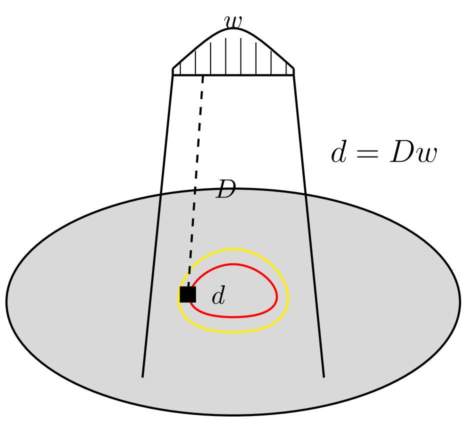


Figure 6: Schematic of an intensity profile (the weights included in the w vector) and a dose distribution described by the d vector. An arrow is drawn (representing an element of the D matrix) indicating the influence of a beam element on a dose element.

It is more instructive to write the dose calculation equation as a summation, where the dose in the ith voxel can be expressed as

$$d_i = \sum_{j=1}^{M} D_{ij} w_j. \tag{1}$$

Here j is the beamlet index and there are M beamlets over all beams used. Note that D_{ij} is the dose delivered to the ith voxel by the jth beamlet for a unit beam weight. A unit of beam weight may be a unit of beam fluence, energy fluence, opening density, exposure time, or any quantity that the D_{ij} matrix is normalized with respect to.

The Objective Function

An optimization algorithm needs an objective function to measure how far the result currently is from the desired solution. For a dose optimization problem, the objective function needs to express an overall voxel-by-voxel deviation from the desired dose, including the importance, or penalty, for each of those deviations. Deviations need not include a subtraction of desired dose from actual dose, but this is a useful and obvious measure. In fact, a quadratic objective function—including a weighted sum of the squared difference between desired and actual dose—is often used. As mentioned above, there may be several goal doses that apply to different voxels, and these goal doses act as the desired dose. If we denote the goal dose for the ith voxel as a prescript dose, d_i^{pres}, and the associated penalty for deviating from that goal as s_i, then the objective function can be defined as

$$F = \sum_{i=1}^{N} s_i \left(d_i - d_i^{pres}\right)^2. \tag{2}$$

Here there are N voxels in which dose is to be considered throughout the dose matrix and corresponding to both targets and sensitive structures. The penalty values are typically user-defined by type and target/structure. If a voxel i is in a given structure and its dose value d_i exceeds the maximum dose goal for that structure, d_i^{pres} in Equation 2 will be the maximum dose goal for that structure. Similarly for structure voxels with dose beneath a minimum dose goal, d_i^{pres} is the minimum dose.

Iterative Optimization of Beamlet Weights

In order to drive the dose to a solution that better meets the pre-defined optimization goals, the most efficient tactic is to alter beamlet weights in some deterministic way. One such method is a so-called gradient technique, where the idea is to reduce the objective function value via calculated changes in the beamlet weights until its rate of change is zero. That is, the objective function is minimized. This concept is illustrated in Figure 7, where as iterations continue, the gradient of the objective function change decreases until it reaches a plateau and the user (or system) would cease the optimization.

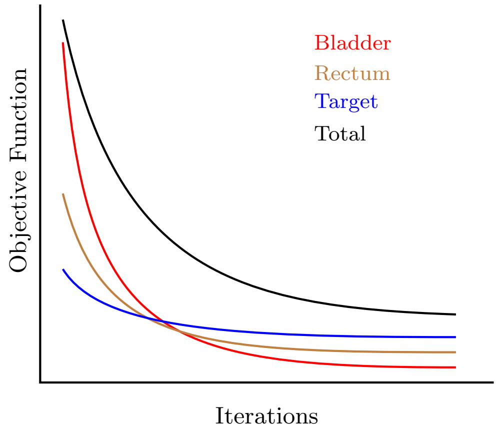


Figure 7: As the iterations progress, the objective function is reevaluated for each structure that has an objective assigned to it. The gradient descent process aims to reduce the difference between the desired dose objective and the actual dose value with each iteration, resulting in the objective function described in Equation 2 approaching zero as all objectives are met. Visualization of each individual objective function allows the user to determine which structure is contributing the most to the objective function, thus is farthest away from the desired dose.

According to the commonly used quasi-Newton gradient technique, the calculated changes in beamlet weight can be arrived at via the derivative and second derivative of the objective function with respect to that particular weight. The change in weight of beamlet j for a given iteration will be

$$\Delta w_j = -\alpha \left(\frac{dF}{dw_j} \Big/ \frac{d^2F}{dw_j^2} \right). \tag{3}$$

The factor α is a damping term to ensure convergence to a solution. The proviso must be imposed that the adjusted weight cannot be less than zero (if it would become so, it is set to zero). Now using Equation (6.3), the change in weight becomes

$$\Delta w_j = -\alpha \frac{\sum_{i=1}^{N} s_i \left(d_i - d_i^{pres}\right) D_{ij}}{\sum_{i=1}^{N} s_i D_{ij}^2}. \tag{4}$$

Note that the weight change for beamlet j is essentially a weighted average of the dose-differences for all voxels influenced by that beamlet, where the weights in the average are dependent on the penalty value appropriate to that voxel and on the dose per unit beamlet weight in that voxel. Thus, as should intuitively be the case, the influence of a particular voxel's dose-deviation on a given beamlet weight is larger when the dose per unit beamlet weight is higher for that voxel.

3.2 matRad Workflow

`matRad` is an open-source dose calculation and optimization toolkit developed entirely in MATLAB, explicitly designed for three-dimensional intensity-modulated radiation therapy (IMRT) research and educational purposes [1]. The software features a highly modular and sequential design that mirrors a clinical treatment planning environment. After importing patient anatomical models—either from external DICOM data or utilizing pre-loaded datasets like the CORT database—users can systematically compute and evaluate plans. The standard `matRad` workflow comprises four primary sequential processes:

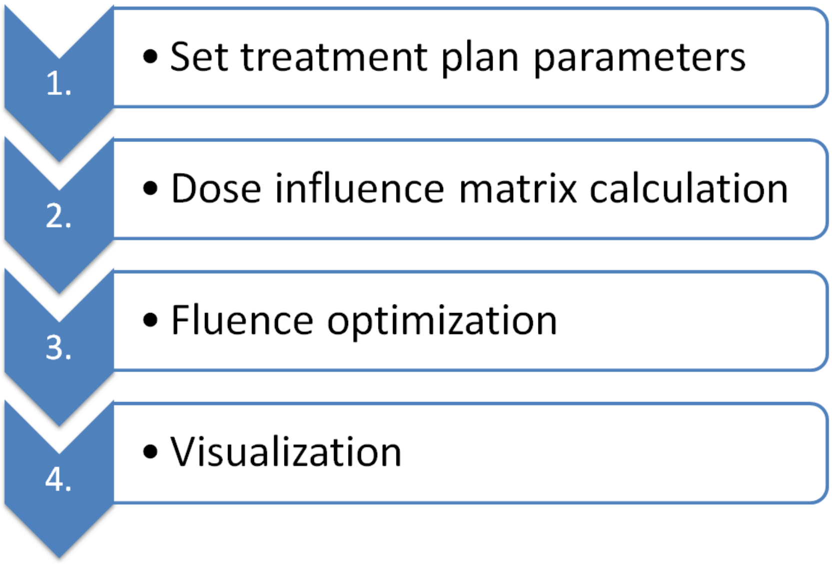


Figure 8: The modular workflow of `matRad` illustrating the sequential progression from plan parameterization to final visualization.

3.2.1 Set treatment plan parameters

Before dose calculation and inverse planning can commence, general treatment plan parameters must be defined. In `matRad`, this initialization corresponds to the first section of the script or the initial interactive settings in the GUI. These parameters are stored within the `pln` struct, which serves as a central repository for plan meta-information. Key parameters include the radiation modality (`pln.radiationMode` for photons, protons, or carbon ions), the specific treatment machine model (`pln.machine`), and the number of fractions (`pln.numOfFractions`), which is particularly critical for biological optimization in particle therapy. Additionally, the `pln` struct is organized into substructures: `pln.propStf` defines steering parameters such as gantry and couch angles (`pln.propStf.gantryAngles`, `pln.propStf.couchAngles`), bixel width, and the isocenter; `pln.propOpt` controls optimization settings including the type of biological optimization (e.g., physical dose, effect-based, or RBE-weighted dose); and `pln.propDoseCalc` allows for the specification of the dose calculation grid resolution.

3.2.2 Dose influence matrix calculation

The dose influence matrix calculation in `matRad` involves two main steps: determining the irradiation geometry and pre-computing the dose influence matrices (D_{ij}) for inverse planning. The irradiation geometry is established using a ray and bixel concept, where the target volume is covered by equidistant rays from a virtual source. For photons, `matRad` employs a singular value decomposed pencil beam algorithm, assuming a uniform primary fluence and modeling the finite source size by convolving the primary fluence with a Gaussian filter. For particles (protons and carbon ions), a conventional pencil beam model is used, combining tabulated depth-dose curves with depth-dependent lateral beam broadening. The resulting dose contributions from each bixel or pencil beam to every voxel are stored in the `dij` struct using a sparse matrix format to optimize memory usage.

3.2.3 Fluence optimization

The core task of fluence optimization is to determine the set of bixel or spot weights that yields a dose distribution best matching the clinical goals. `matRad` formulates this as a mathematical optimization problem, minimizing a weighted sum of objective functions subject to constraints. Supported objectives include squared overdosage and underdosage, mean dose, and equivalent uniform dose (EUD), as well as dose-volume histogram (DVH) constraints. The software utilizes the IPOPT (Interior Point OPTimizer) package for solving these large-scale non-linear optimization problems. The framework supports optimization based on physical dose, as well as biological effect and RBE-weighted dose for particle therapy, allowing for comprehensive treatment planning flexibility.

Table 1: Supported objective and constraint functions by matRad with dose d_i in voxel i, prescribed dose $\hat{d}$, dose $\tilde{d}$ at prescribed volume, structure S, number of voxels in structure N_S, equivalent uniform dose (EUD) exponent a, Heaviside function $\Theta(x)$, and log-sum-exp function parameter $\kappa = 10^{-3}$. Min/max mean and min/max EUD constraints are not explicitly stated but are supported by matRad. [1]

Objectives		Constraints	
$f_{\text{sq deviation}}$	$= \frac{1}{N_S}\sum_{i\in S}(d_i - \hat{d})^2$		
$f_{\text{sq under dosage}}$	$= \frac{1}{N_S}\sum_{i\in S}\Theta(\hat{d} - d_i)(d_i - \hat{d})^2$	$c_{\text{min dose}}$	$= d_{\text{min}} - \kappa\log(\sum_{i\in S} e^{\frac{d_{\text{min}} - d_i}{\kappa}})$
$f_{\text{sq over dosage}}$	$= \frac{1}{N_S}\sum_{i\in S}\Theta(d_i - \hat{d})(d_i - \hat{d})^2$	$c_{\text{max dose}}$	$= d_{\text{max}} + \kappa\log(\sum_{i\in S} e^{\frac{d_i - d_{\text{max}}}{\kappa}})$
f_{mean}	$= \frac{1}{N_S}\sum_{i\in S} d_i$	c_{mean}	$= \frac{1}{N_S}\sum_{i\in S} d_i$
f_{EUD}	$= \left(\frac{1}{N_S}\sum_{i\in S} d_i^a\right)^{\frac{1}{a}}$	c_{EUD}	$= \left(\frac{1}{N_S}\sum_{i\in S} d_i^a\right)^{\frac{1}{a}}$
$f_{\text{min DVH}}$	$= \frac{1}{N_S}\sum_{i\in S}\Theta(\hat{d} - d_i)\Theta(d_i - \tilde{d})(d_i - \hat{d})^2$	$c_{\text{min DVH}}$	$= \frac{1}{N_S}\sum_{i\in S}\Theta(\hat{d} - d_i)$
$f_{\text{max DVH}}$	$= \frac{1}{N_S}\sum_{i\in S}\Theta(d_i - \hat{d})\Theta(\tilde{d} - d_i)(d_i - \hat{d})^2$	$c_{\text{max DVH}}$	$= \frac{1}{N_S}\sum_{i\in S}\Theta(d_i - \hat{d})$

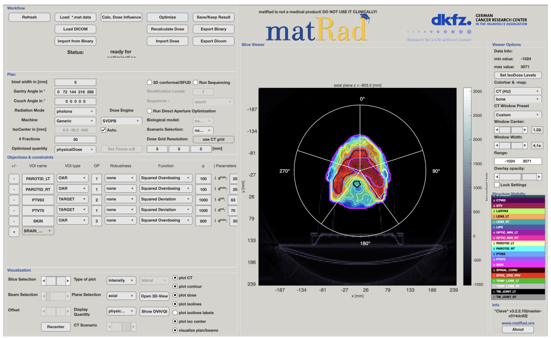
3.2.4 Visualization

Visualization in `matRad` is primarily handled through its Graphical User Interface (GUI), which can be launched via the `matRadGUI` command. The GUI automatically detects the current state of the treatment plan in the workspace and displays relevant data, such as patient anatomy and beam geometry. Once the optimization process is complete, the calculated 3D dose distribution is visualized overlaid on the CT data. Users can interactively explore the plan through axial, sagittal, and coronal views, evaluate dose-volume histograms (DVHs), and analyze dose profiles (e.g., longitudinal profiles along the central beam axis) to assess plan quality.

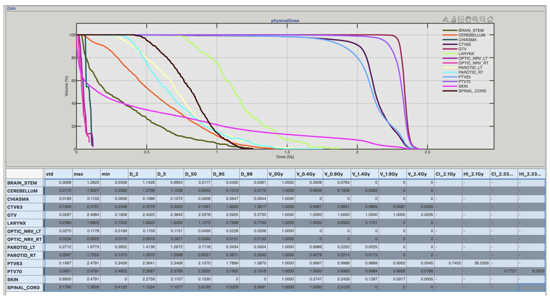
4 Observation

4.1 Head and Neck Planning

4.1.1 5 Beam Planning



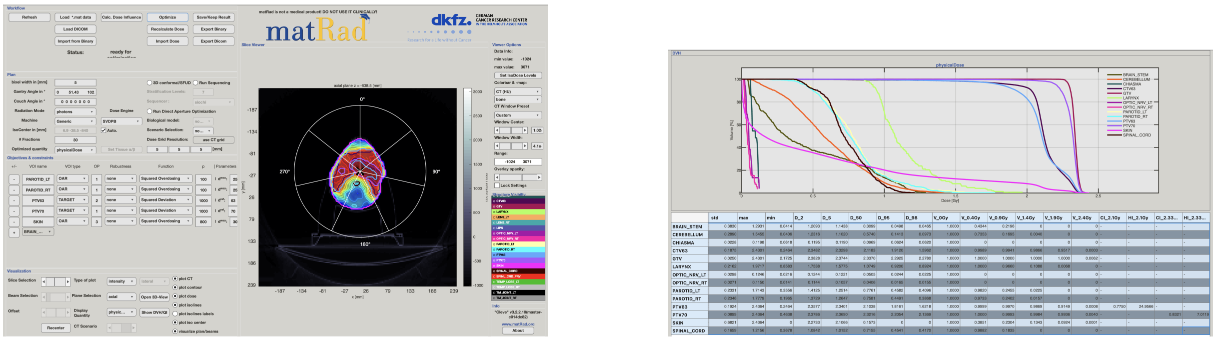
(a) 5 Beam Planning



(b) DVH of 5 Beam Planning

Figure 9: 5 Beam Planning and its DVH

4.1.2 7 Beam Planning

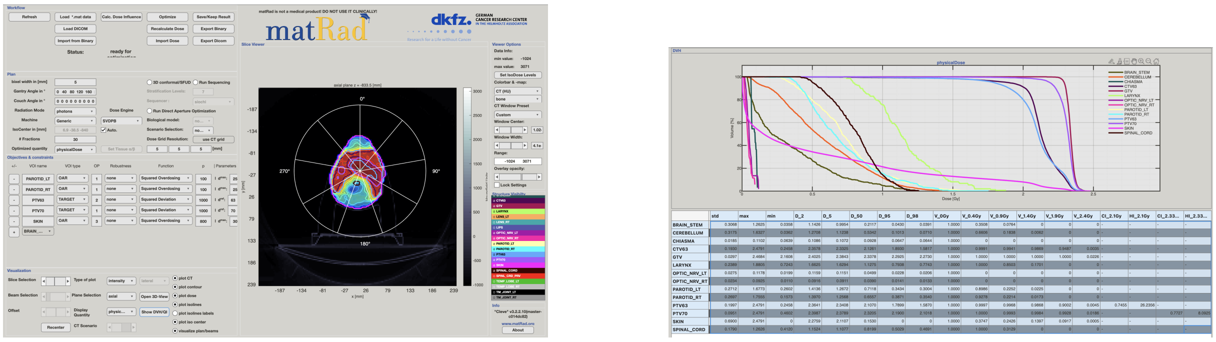


(a) 7 Beam Planning

(b) DVH of 7 Beam Planning

Figure 10: 7 Beam Planning and its DVH

4.1.3 9 Beam Planning



(a) 9 Beam Planning

(b) DVH of 9 Beam Planning

Figure 11: 9 Beam Planning and its DVH

	mean	std	max	min	D_2	D_5	D_50	D_95	D_98	V_0	V_0.4	V_0.9	V_1.4	V_1.9	V_2.4	CI_2.1	HI_2.1	CI_2.33	HI_2.33
5 Beams																			
Brain_stem	0.3376	0.3068	1.2625	0.0358	1.1426	0.9954	0.2117	0.043	0.0391	1	0.3508	0.0764	0	0	0	-	-	-	-
Cerebellum	0.5797	0.3175	1.6327	0.0362	1.2708	1.1238	0.5342	0.1013	0.071	1	0.6606	0.1838	0.0062	0	0	-	-	-	-
Chiasma	0.0855	0.0185	0.1102	0.0639	0.1086	0.1072	0.0928	0.0647	0.0644	1	0	0	0	0	0	-	-	-	-
CTV63	2.1206	0.193	2.4791	0.2458	2.3578	2.3325	2.1261	1.893	1.5817	1	0.9991	0.9941	0.9869	0.9487	0.0035	-	-	-	-
GTV	2.3379	0.0297	2.4684	2.1608	2.4025	2.3843	2.3378	2.2925	2.273	1	1	1	1	1	0.0226	-	-	-	-
Larynx	1.153	0.2389	1.8805	0.7243	1.6625	1.6294	1.1275	0.7938	0.7743	1	1	0.8503	0.1701	0	0	-	-	-	-
Optic nerve LT	0.0553	0.0275	0.1178	0.0199	0.1159	0.1151	0.0499	0.0228	0.0206	1	0	0	0	0	0	-	-	-	-
Optic nerve RT	0.0438	0.0234	0.0925	0.011	0.0916	0.0911	0.039	0.0141	0.0133	1	0	0	0	0	0	-	-	-	-
Parotid LT	0.7402	0.2712	1.6773	0.2602	1.4136	1.2672	0.7118	0.3434	0.3004	1	0.8986	0.2252	0.0225	0	0	-	-	-	-
Parotid RT	0.7117	0.2697	1.7555	0.1573	1.397	1.2568	0.6557	0.3871	0.354	1	0.9278	0.2214	0.0173	0	0	-	-	-	-
PTV63	2.0945	0.1997	2.4791	0.2458	2.3641	2.3408	2.107	1.7899	1.587	1	0.9997	0.9968	0.9868	0.9002	0.0045	0.7455	26.2356	-	-
PTV70	2.3025	0.0951	2.4791	0.4602	2.3987	2.3789	2.3205	2.19	2.1018	1	1	0.9993	0.9984	0.9928	0.0186	-	-	0.7727	8.0925
skin	0.5229	0.69	2.4791	0	2.2759	2.1107	0.153	0	0	1	0.3747	0.2426	0.1397	0.0917	0.0005	-	-	-	-
Spinal cord	0.8089	0.179	1.2626	0.412	1.1524	1.1077	0.8199	0.5029	0.4691	1	1	0.3129	0	0	0	-	-	-	-
7 Beams																			
Brain_stem	0.4635	0.383	1.2931	0.0414	1.2093	1.1438	0.3099	0.0498	0.0465	1	0.4344	0.2196	0	0	0	-	-	-	-
Cerebellum	0.6024	0.289	1.5455	0.0406	1.2316	1.102	0.574	0.1413	0.0973	1	0.7353	0.1695	0.004	0	0	-	-	-	-
Chiasma	0.0925	0.0228	0.1198	0.0618	0.1195	0.119	0.0969	0.0624	0.062	1	0	0	0	0	0	-	-	-	-
CTV63	2.117	0.1875	2.4301	0.2464	2.3482	2.3298	2.1183	1.912	1.5962	1	0.9989	0.9941	0.9866	0.9517	0.0003	-	-	-	-
GTV	2.3361	0.025	2.4301	2.1725	2.3828	2.3744	2.337	2.2925	2.278	1	1	1	1	1	0.0062	-	-	-	-
Larynx	1.1438	0.2162	1.9717	0.8583	1.7538	1.5775	1.0749	0.92	0.8924	1	1	0.966	0.1088	0.0068	0	-	-	-	-
Optic nerve LT	0.0576	0.0298	0.1246	0.0216	0.1244	0.1221	0.0505	0.0244	0.0225	1	0	0	0	0	0	-	-	-	-
Optic nerve RT	0.0466	0.0271	0.115	0.0141	0.1144	0.1057	0.0406	0.0165	0.0155	1	0	0	0	0	0	-	-	-	-
Parotid LT	0.7937	0.2331	1.7143	0.3556	1.4125	1.2514	0.7761	0.4582	0.4096	1	0.982	0.2455	0.0225	0	0	-	-	-	-
Parotid RT	0.7778	0.2346	1.7779	0.1965	1.3729	1.2647	0.7581	0.4491	0.3868	1	0.9733	0.2402	0.0157	0	0	-	-	-	-
PTV63	2.0972	0.1924	2.4364	0.2464	2.3577	2.3401	2.1038	1.8161	1.6218	1	0.9999	0.997	0.9869	0.9149	0.0008	0.775	24.9566	-	-
PTV70	2.3037	0.0899	2.4364	0.4638	2.3786	2.369	2.3216	2.2054	2.1369	1	1	0.9993	0.9984	0.9936	0.004	-	-	0.8321	7.0119
skin	0.518	0.6821	2.4364	0	2.2733	2.1066	0.1573	0	0	1	0.3851	0.2304	0.1343	0.0924	0.0001	-	-	-	-
Spinal cord	0.739	0.1659	1.2156	0.3678	1.0842	1.0152	0.7155	0.4541	0.417	1	0.9882	0.1835	0	0	0	-	-	-	-
9 Beams																			
Brain_stem	0.3987	0.3401	1.0828	0.042	1.052	1.0371	0.2648	0.0493	0.0456	1	0.3699	0.1742	0	0	0	-	-	-	-
Cerebellum	0.6124	0.3011	1.49	0.0421	1.1679	1.0831	0.6054	0.1497	0.1028	1	0.6957	0.2065	0.0018	0	0	-	-	-	-
Chiasma	0.0987	0.0285	0.1362	0.0648	0.1361	0.1356	0.1073	0.0667	0.0653	1	0	0	0	0	0	-	-	-	-
CTV63	2.1147	0.1889	2.4845	0.2569	2.3539	2.335	2.1138	1.9076	1.5729	1	0.999	0.9943	0.9861	0.951	0.0022	-	-	-	-
GTV	2.3366	0.0249	2.441	2.1479	2.389	2.3763	2.336	2.297	2.283	1	1	1	1	1	0.0092	-	-	-	-
Larynx	1.1949	0.1527	1.7896	0.9524	1.6173	1.4945	1.1613	1.0181	0.9971	1	1	1	0.0952	0	0	-	-	-	-
Optic nerve LT	0.0574	0.0299	0.1319	0.0211	0.1278	0.1225	0.0475	0.0243	0.0221	1	0	0	0	0	0	-	-	-	-
Optic nerve RT	0.0532	0.0338	0.1418	0.0182	0.1367	0.1284	0.0416	0.0198	0.0192	1	0	0	0	0	0	-	-	-	-
Parotid LT	0.7796	0.23	1.6611	0.382	1.3855	1.2295	0.7569	0.4516	0.4164	1	0.991	0.2477	0.0203	0	0	-	-	-	-
Parotid RT	0.7727	0.2341	1.6631	0.171	1.3213	1.2323	0.7724	0.416	0.3354	1	0.9576	0.2402	0.0126	0	0	-	-	-	-
PTV63	2.0964	0.1936	2.4845	0.2569	2.3615	2.3432	2.1023	1.8155	1.6118	1	0.9999	0.9967	0.9866	0.913	0.0019	0.7797	25.1247	-	-
PTV70	2.3054	0.0919	2.4845	0.4657	2.3846	2.3708	2.3247	2.1987	2.1275	1	1	0.9993	0.9982	0.9939	0.0074	-	-	0.8516	7.3764
skin	0.5218	0.678	2.4845	0	2.2745	2.1046	0.1557	0	0	1	0.392	0.2326	0.1303	0.0916	0.0002	-	-	-	-
Spinal cord	0.8612	0.1978	1.274	0.4585	1.2031	1.1639	0.8742	0.5402	0.5007	1	1	0.4541	0	0	0	-	-	-	-

Figure 12: Dose Volume Histogram (DVH) for Head and Neck Planning

QUANTEC-Based Evaluation of Treatment Plans (SIB Technique)

The treatment plan was generated using a Simultaneous Integrated Boost (SIB) technique, delivering different prescription doses to multiple target volumes in 30 fractions:

- PTV70: 70 Gy in 30 fractions (≈ 2.33 Gy/#)
- PTV63: 63 Gy in 30 fractions (≈ 2.10 Gy/#)

The DVH values represent dose per fraction (Gy/#) and were multiplied by 30 to obtain total dose in Gy.

For organs at risk (OARs), evaluation was performed using absolute dose values and compared directly with QUANTEC guidelines. For serial organs such as the spinal cord, $D_{2\%}$ was used as a surrogate for $D_{\max}$.

Spinal Cord

QUANTEC recommends $D_{\max} < 45$ Gy.

- 5 beams: $D_2 = 34.57$ Gy
- 7 beams: $D_2 = 32.53$ Gy
- 9 beams: $D_2 = 36.09$ Gy

All plans satisfy the constraint. The 7-beam configuration provides the best sparing of the spinal cord.

Brainstem

QUANTEC recommends $D_{\max} < 54$ Gy.

- 5 beams: $D_2 = 34.28$ Gy
- 7 beams: $D_2 = 36.28$ Gy
- 9 beams: $D_2 = 31.56$ Gy

All plans are within tolerance, with the 9-beam configuration providing the lowest dose.

Parotid Glands

QUANTEC recommends mean dose $D_{\text{mean}} < 26$ Gy.

- Left Parotid:
  - 5 beams: 22.21 Gy
  - 7 beams: 23.81 Gy
  - 9 beams: 23.39 Gy
- Right Parotid:
  - 5 beams: 21.35 Gy
  - 7 beams: 23.33 Gy
  - 9 beams: 23.18 Gy

All plans satisfy QUANTEC criteria. The 5-beam configuration provides superior parotid sparing and is preferable for reducing xerostomia risk.

Optic Nerves and Chiasm

QUANTEC recommends $D_{\max} < 54$ Gy.

- All plans: $D_2 < 5$ Gy

These structures are well below tolerance in all configurations.

Larynx

QUANTEC recommends mean dose $D_{\text{mean}} < 44$ Gy.

- 5 beams: 34.59 Gy
- 7 beams: 34.31 Gy
- 9 beams: 35.85 Gy

All plans meet the constraint, with the 7-beam configuration showing slightly improved sparing.

Target Coverage

Target volumes were evaluated relative to their respective prescriptions:

- PTV70 (70 Gy prescription):
  - D_{98}: 63.05 Gy (5 beams), 64.11 Gy (7 beams), 63.83 Gy (9 beams)
- PTV63 (63 Gy prescription):
  - D_{98}: $\approx$ 47.5–48.6 Gy across all plans

All plans achieve acceptable coverage of both target volumes.

Conclusion

All treatment plans satisfy QUANTEC dose constraints for organs at risk.

- The 7-beam plan provides the best overall balance between target coverage and spinal cord sparing.
- The 5-beam plan achieves superior parotid sparing and may be preferred when minimizing xerostomia is prioritized.
- The 9-beam plan increases spinal cord dose and low-dose spread without significant clinical advantage.

Thus, the 7-beam configuration is considered the most optimal plan under QUANTEC-based evaluation.

5 Conclusion

This experiment provided a comprehensive familiarization with the operations and workflow of a clinical Treatment Planning System (TPS). Through practical, hands-on experience, the fundamental processes of radiation therapy planning were successfully executed, including the importation of patient CT datasets, delineation of target volumes (PTVs) and organs at risk (OARs), beam configuration, dose calculation, and the final quantitative evaluation of the treatment plans.

Specifically, the formulation and optimization of an Intensity Modulated Radiation Therapy (IMRT) plan using a Simultaneous Integrated Boost (SIB) technique for a Head and Neck case was evaluated. The targets, PTV70 and PTV63, were prescribed 70 Gy and 63 Gy respectively over 30 fractions. By comparing 5-beam, 7-beam, and 9-beam configurations, significant insights into the dosimetric trade-offs of beam multiplicity were gained.

Quantitative evaluation using Dose Volume Histograms (DVHs) and QUANTEC guidelines verified that all configurations successfully maintained doses to critical structures (spinal cord, brainstem, parotid glands, optic nerves/chiasm, and larynx) within safe clinical tolerances while achieving acceptable target coverage (D_{98}). The comparative analysis highlighted that simply increasing the number of beams does not unilaterally improve a plan:

- The **7-beam configuration** provided the optimal overarching balance, yielding excellent target coverage while effectively suppressing the maximum dose to the spinal cord.
- The **5-beam configuration** resulted in the most superior sparing of the parotid glands, making it a viable alternative if prioritizing the reduction of xerostomia risks over other factors.
- The **9-beam configuration** failed to offer a meaningful clinical advantage, instead leading to an increased low-dose wash to surrounding healthy tissues and slightly higher spinal cord doses.

Ultimately, this exercise distinctly highlighted the intricate optimization challenges inherent to modern radiotherapy. It demonstrated the paramount importance of critical plan evaluation and justified compromise to deliver maximum therapeutic effect while assuring patient safety.

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